

Problem Set – More on Pass By Value Functions. Create an IPO for each of the problems below. Save the document with the IPO's and then upload to Blackboard. Next write code for the problems. Then upload the .cpp files to Blackboard.

1. Allow the user to enter a quantity and price, use `ctrl+z` to stop. Use a function to compute the total (quantity times price). The function should be passed the quantity and price and then return the total. Use another function to compute 10% discount if the total is over \$10,000.00 and 5% for any amount equal to or lower than \$10,000.00. The second function should receive the total, check which discount rate to charge, compute discount amount and then compute the discount total (total – discount amount). It should return the discount total. Display total and discount total. Sum total and discount total and display at the end.

Input	Process	Output
• Quantity • Price	<i>computeTotal(Quantity and Price)</i> • Declare Total • Calculate Total by finding the product of Quantity and Price • Return Total	<i>For Each Entry</i> • Total • Discounted Total
	<i>computeDiscounts(Total)</i> • Declare Discount and Discounted Total • If Total is greater than 10,000, calculate Discount by finding the product of 10% (.10) and Total ○ Otherwise, calculate Discount by finding the product of 5% (.05) • Calculate Discounted Total by finding the difference of Total and Discount • Return Discounted Total	<i>After All Entries</i> • Sum of Totals • Sum of Discounted Totals
	<i>Main()</i> • Declare Quantity, Price, Total, and Discounted Total • Declare Sum of Totals and Sum of Discounted Totals, set at 0 • Assign the input for Quantity and Price as their variables • While not EOF • <u>Call <i>computeTotal</i></u> and set as Total • <u>Call <i>computeDiscounts</i></u> and set as Discounted Total • Add Total to Sum of Totals • Add Discounted Total to Sum of Discounted Totals • Display Total and Discounted Total • Ask for new entry for Quantity and Price • End Loop	

	• Display Sum of Totals and Sum of Discounted Totals	
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2. Enter players last name, number of hits and at bats at the keyboard, use ctl+z to stop. Use a function to compute batting average. Pass the hits and at bats to the function. The function should return batting average (at bats / number of hits). Display last name and batting average. Give a count of the number of players entered and display the count after the loop.

Input	Process	Output
• Player's Last Name • # of Hits • # of At Bats	<i>computeBattingAvg(# of Hits and # of At Bats)</i> • Declare Batting Average • Calculate Batting Average by finding the quotient of # of At Bats and # of Hits • Return Batting Average	<i>For Each Entry</i> • Player's Last Name • Batting Average
	<i>Main()</i> • Declare Player's Last Name, # of Hits, # of At Bats, and Batting Average • Declare # of Players, set at 0 • Assign the input for Player's Last Name, # of Hits, and # of At Bats as their variables • While not EOF • Call <i>computeBattingAvg</i> and set as Batting Average • Add 1 to # of Players • Display Player's Last Name and Batting Average • Ask for new entry for Player's Last Name, # of Hits, and # of At Bats • End Loop • Display # of Players	<i>After All Entries</i> • # of Players

3. Enter the destination city, miles travelled and gallons used for a trip, use ctl+z to stop. Use a function to compute miles per gallon. Pass miles travelled and gallons used to the function. The function should return miles per gallon. Use another function to compute gas cost. Pass to this function gallons used. Each gallon costs \$3.50. Compute and return the cost. Display destination city, miles per gallon and cost of gas. Sum and display the total cost of gas.

Input	Process	Output
• Destination City • Miles Travelled • Gallons Used	<i>computeMpg(Miles Travelled and Gallons Used)</i> • Declare Mpg • Calculate Mpg by finding the quotient of Miles Travelled and Gallons Used • Return Mpg	<i>For Each Entry</i> • Destination City • Mpg • Cost of Gas
	<i>computeGasCost(Gallons Used)</i> • Declare Cost of Gas	<i>After All Entries</i>

	• Calculate Cost of Gas by finding the product of Gallons Used and \$3.50 • Return Cost of Gas	• Total Cost of Gas
	<i>Main()</i> • Declare Destination City, Miles Travelled, Gallons Used, Mpg, and Cost of Gas • Declare Total Cost of Gas, set at 0 • Assign the input for Destination City, Miles Travelled, Gallons Used as their variables • While not EOF • <u>Call <i>computeMpg</i></u> and set as Mpg • <u>Call <i>computeGasCost</i></u> and set as Cost of Gas • Add Cost of Gas to Total Cost of Gas • Display Destination City, Mpg, and Cost of Gas • Ask for new entry for • End Loop • Display Total Cost of Gas	

4. Allow the employee to enter last name, job code and hours worked, use ctl+z to stop. Use a function to determine the pay rate. Pass to this function the job code and it should return rate of pay. Use the following rates based on Job code: L is \$25/hr, A is \$30/hr and J is \$50/hr for respective pay rates. Write another function to determine the gross pay. Pass to this function the hours worked and pay rate and return gross pay. Give time and a half for overtime. Display last name and gross pay. Sum and display total of all gross pay.

Input	Process	Output
• Employee's Last Name • Job Code • Hours Worked	<i>computePayRate(Job Code)</i> • Declare Pay Rate • If Job Code equals L, set Pay Rate at \$25 ○ If Job Code equals A, set Pay Rate at \$30 ○ If Job Code equals J, set Pay Rate at \$50 ○ Otherwise, End Program • Return Pay Rate	For Each Entry • Employee's Last Name • Gross Pay
	<i>computeGrossPay(Hours Worked and Pay Rate)</i> • Declare Gross Pay and Overtime • If Hours Worked is greater than 40, calculate Overtime by finding the difference of Hours and 40, then calculate Gross Pay by finding the sum of 40 * Pay Rate and the product of Overtime and Pay Rate * 1.5 ○ Otherwise, calculate Gross Pay by finding the product of Hours and Pay Rate • Return Gross Pay	After All Entry • Total Gross Pay
	<i>Main()</i>	

	• Declare Employee's Last Name, Job Code, Hours Worked and Gross Pay • Declare Total Gross Pay, set at 0 • Assign the input for Employee's Last Name, Job Code, and Hours Worked as their variables • While not EOF • <u>Call <i>computePayRate</i></u> and set a Pay Rate • <u>Call <i>computeGrossPay</i></u> and set as Gross Pay • Add Gross Pay to Total Gross Pay • Display Employee's Last Name and Gross Pay • Ask for new entry for Employee's Last Name, Job Code, and Hours Worked • End Loop • Display Total Gross Pay	
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5. Allow the user to enter student last name, credit hours and district code, use ctrl+z to stop. Use a function to compute tuition owed. First, write a function to determine the cost per credit hour. Charge In district (code of I) \$250 per credit hour. Out of district (code of O) is \$550 per credit hour. Write another function to compute tuition cost. Display student name and tuition cost. Sum and display total of all tuition costs.

Input	Process	Output
• Student's Last Name • Credit Hours • District code	<i>computeCreditHourRate(District Code)</i> • Declare Credit Hour Rate • If District Code equals I (In-District), set Credit Hour Rate at \$250 ○ If District Code equals O (Out of District), set Credit Hour Rate at \$550 ○ Otherwise, End Program • Return Credit Hour Rate	For Each Entry • Student's Last Name • Tuition Cost
	<i>computeTuition(Credit Hours and Credit Hour Rate)</i> • Declare Tuition • Calculate Tuition by finding the product of Credit Hours and Credit Hour Rate • Return Tuition	After All Entry • Total Tuition Cost
	<i>Main ()</i> • Declare Student's Last Name, Credit Hours, District Code, Credit Hour Rate, and Tuition • Declare Total Tuition Cost, set at 0 • Assign the input for Student's Last Name, Credit Hours, and District Code as their variables • While not EOF	

	• Call <u><i>computeCreditHourRate</i></u> and set as Credit Hour Rate • Call <u><i>computeTuition</i></u> and set as Tuition • Add Tuition to Total Tuition • Display Student's Last Name and Tuition • Ask for new entry for Student's Last Name, Credit Hours, and District Code • End Loop • Display Total Tuition Cost	
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